

A Transparent Monte Carlo Report

Elite Study Career vs. High-Level Professional Soccer

The practical question

If a young person starts early and works seriously, which path is statistically safer: building an elite study-based career, or trying to become a high-level professional footballer?

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Frozen run used in this version. The report keeps the current simulation outputs unchanged: $N = 300,000$, seed 42, model version 1.0, with a 60,000-observation machine-learning audit sample. The text has been rewritten for a more natural, more defensible report style; the numerical content has not been softened.

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1 The honest answer, before the mathematics

This report started from a question that is simple, but emotionally loaded: should a talented child protect school first, or can football rationally become the main plan?

The model does not insult football. It does not say that dreams are foolish. It says something narrower and more useful: **high-level football is a tiny-capacity market with a short biological selection window**. School is also hard, but it has more routes, more second chances, and more ways to convert discipline into a strong life outcome.

Main conclusion. In the frozen run used here, the simulated probability of an elite study outcome is **10.839%** (about 1 in 9.2). The probability of any professional soccer contract is **0.260%** (about 1 in 384). The probability of high-level professional soccer is **0.088%** (about 1 in 1,136). That makes the elite-study route about **123.2 times** more likely than high-level professional soccer for the default goal-committed cohort.

The right interpretation is not: “never play football.” The right interpretation is: **do not burn school unless football has already been validated by serious external selection**. A young person can train seriously, compete seriously, and still protect the academic base plan. That dual-track strategy is what the numbers support.

A useful way to say it is this: ambition deserves respect, but risk deserves measurement.

لولا المشقة ساد الناس كلهم
الجود يفقر والإقدام قتال

Without hardship, everyone would become a master; generosity can make one poor, and courage can be fatal.

The Arabic lines are kept deliberately. They are not decoration; they express the same idea as the model: ambition has a price, and serious choices deserve serious measurement.

2 What exactly is being compared?

Let i index a simulated child. The model compares three binary outcomes:

$$\begin{aligned} Y_i^A &= 1 && \text{if child } i \text{ reaches an elite academic/professional study outcome,} && (1) \\ Y_i^{F,pro} &= 1 && \text{if child } i \text{ earns a professional soccer contract,} && (2) \\ Y_i^{F,elite} &= 1 && \text{if child } i \text{ reaches high-level professional soccer.} && (3) \end{aligned}$$

The academic outcome is deliberately stricter than simply attending university. It means entering and completing a selective pathway that can plausibly lead to elite engineering, medicine, research, applied science, or a comparable technical career.

The football side is split into two levels because a lower-division professional contract and a stable high-level career are not the same thing. They should not be mixed.

Both routes are conditioned on early serious commitment:

$$G_i = 1.$$

This means the child is not a random child with no interest. The child starts early and receives

some preparation or training. Even inside that committed cohort, the model still allows differences in ability, discipline, family resources, school or club environment, health, injuries, and luck.

3 External anchors used for calibration

There is no public dataset that follows every child from age six to elite engineer, doctor, or professional footballer. The honest approach is therefore not to pretend to have a magic empirical number. The model uses public anchors to set the order of magnitude, then makes the assumptions explicit.

Area	Anchor used in the model	Source
Professional soccer capacity	FIFA reported 128,694 professional male footballers at 3,986 clubs in 135 countries in its Professional Football Report 2023.	FIFA 2023
Academy-to-elite soccer	A University of Essex summary of a recent academy follow-up reported that about 4% of talented teen academy prospects reached top-tier professional football, while about 6% played in lower leagues.	Essex 2024
Injury risk	A systematic review estimated 8.1 injuries per 1,000 exposure hours in professional male football; match incidence was much higher than training incidence.	Lopez-Valenciano et al. 2020
Severe injury definition/rate	The UEFA Elite Club Injury Study classifies injuries causing more than four weeks of absence as severe and reported severe-injury rates at elite-club level.	UEFA ECIS 2019
General education attainment	OECD Education GPS reports that 48% of 25–34 year-olds across OECD countries held a tertiary qualification in 2024; France was 53%.	OECD 2025
Master’s-level attainment	In France, 26% of 25–34 year-olds held a master’s or equivalent degree in 2024, above the OECD average of 16%.	OECD 2025
STEM share	Across OECD countries, 26% of tertiary-educated adults had studied STEM.	OECD 2025
Mathematics top performance	In PISA 2022, 7% of students in France and 9% on average across OECD countries were top performers in mathematics.	OECD PISA 2022
Medicine example	In the United States, AAMC reported 54,699 medical school applicants and 23,440 matriculants in 2025.	AAMC 2025
US soccer example	NCAA reports that 5.9% of US male high-school soccer participants go on to compete at an NCAA level; this is not the same as becoming professional.	NCAA 2025

These anchors do not make the model universal. A child in Morocco, France, Brazil, Spain, Senegal, Germany, Japan, or the United States faces different school systems, football pyramids, scouting networks, costs, and family constraints. The point is transparency: the assumptions are visible and editable.

4 Model structure

4.1 Latent traits

Each simulated child has a vector of standardized latent characteristics:

$$X_i = (C_i, A_i, D_i, R_i, E_i, H_i, P_i, L_i, G_i),$$

where:

- C_i = academic readiness;
- A_i = athletic ability;
- D_i = discipline, consistency, and work capacity;
- R_i = family resources and socioeconomic support;
- E_i = school, coaching, and local environment quality;
- H_i = health resilience;
- P_i = psychological resilience;
- L_i = luck and timing;
- G_i = early goal commitment.

The phrase “low ability” in this model means low current measured readiness inside the simulated committed cohort. It does not mean a person is permanently incapable. Real readiness can change through teaching, sleep, health, stability, feedback, and time.

4.2 Stage-gate probability model

Each path is modelled as a sequence of gates. For pathway d and gate t :

$$p_{d,t,i} = \sigma \left(\alpha_{d,t} + \beta_{d,t}^\top X_i - \Delta_{d,t,i} \right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

A gate is passed using a Bernoulli draw:

$$B_{d,t,i} \sim \text{Bernoulli}(p_{d,t,i}).$$

The final outcome is the product of all required gates:

$$Y_{d,i} = \prod_{t=1}^{T_d} B_{d,t,i}.$$

That product is the key. Even when every single stage looks possible, several stages multiplied together can produce a small final probability.

4.3 Academic gates

The academic pathway uses four gates:

1. **Foundation:** basic skills, literacy, numeracy, and learning habits.
2. **Strong secondary profile:** strong grades, exams, and preparation for selective tracks.
3. **Elite entry:** admission to medicine, elite engineering, or another highly selective technical path.

4. **Completion and elite transition:** finishing the path and entering a high-level professional role.

The strongest academic drivers are academic readiness, discipline, family resources, and environment. Health and psychological resilience matter more strongly when the path becomes long and demanding.

4.4 Football gates

The football pathway also uses four gates:

1. **Regional or academy selection:** being identified as unusually promising.
2. **Academy retention:** surviving the youth filter through adolescence.
3. **Professional contract:** earning a senior professional pathway.
4. **High-level career:** reaching and staying at a high professional level.

Football has harsher late-stage intercepts because it is capacity constrained. There are not many roster slots relative to the number of serious aspiring players.

4.5 Injury and disruption processes

Football is not only a talent filter. It is also a body-risk filter. Severe injuries are modelled as a multi-year count, and career-limiting injury is modelled separately. The model penalizes repeated severe injuries, but it does not assume that every serious injury ends a career.

The model also includes academic life disruption: illness, family instability, forced moves, financial pressure, caring responsibilities, mental-health disruption, or other events that can interrupt education. This is not a measure of intelligence; it is a shock variable.

5 Why the football probability is structurally smaller

The football route is not only hard because football requires talent. It is hard because elite football has a hard capacity ceiling. Let S_F be the number of high-level roster slots and N_F the number of serious aspiring players. Before even discussing individual ability, the average slot probability is bounded roughly by

$$\mathbb{P}(Y^{F,elite} = 1) \lesssim \frac{S_F}{N_F}.$$

Education also has competition, but it has more parallel routes. A strong mathematics student can become an engineer, software specialist, data scientist, doctor, researcher, actuary, quantitative analyst, founder, technical manager, or move into another demanding professional track. A footballer just below the professional threshold does not have fifty equivalent elite football jobs available.

This is the structural reason the model tends to produce

$$\mathbb{P}(Y^A = 1 \mid G = 1) \gg \mathbb{P}(Y^{F,elite} = 1 \mid G = 1).$$

ومن يتَّيَّب صعود الجبال
يعش أبد الدهر بين الحفر

Whoever fears climbing mountains will live forever among pits.

The conclusion is therefore not cowardice. It is controlled courage: climb the mountain, but do not destroy the bridge behind you before the climb has produced evidence.

6 Monte Carlo estimation and audit trail

The code simulates N children and estimates

$$\hat{\mathbb{P}}(Y = 1) = \frac{1}{N} \sum_{i=1}^N Y_i.$$

For each estimated probability $\hat{p}$, the approximate Monte Carlo standard error is

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}},$$

and an approximate 95% interval is

$$\hat{p} \pm 1.96SE(\hat{p}).$$

Run metadata. The frozen output used here has $N = 300,000$, seed 42, model version 1.0, and a 60,000-observation random-forest audit sample. The machine-learning layer is not used to create the probabilities; it is used as an internal coherence check.

This matters. The report is not a vague opinion about school and football. It is a decision model with visible assumptions, visible gates, and a frozen simulation output.

7 Final simulation results

7.1 Headline probabilities

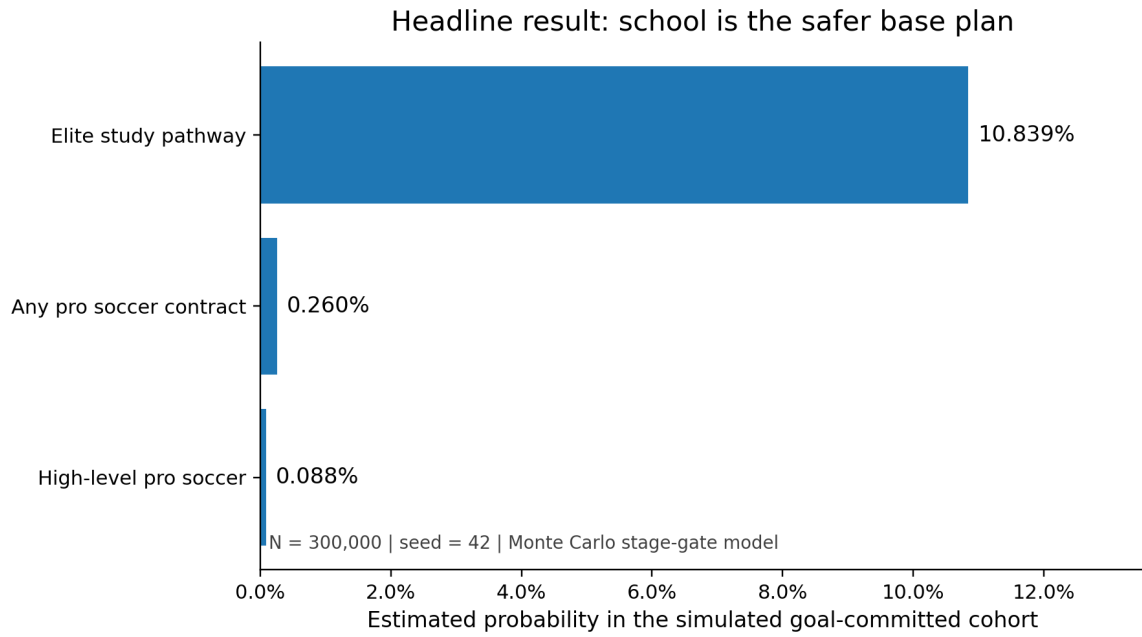


Figure 1: Headline probabilities from the frozen Monte Carlo run. The large gap is the main result.

Under the current calibration, the estimated probability of reaching the elite study pathway is

$$\hat{\mathbb{P}}(Y^A = 1) = 10.839\% \approx 1 \text{ in } 9.2.$$

The estimated probability of any professional soccer contract is

$$\hat{\mathbb{P}}(Y^{F,pro} = 1) = 0.260\% \approx 1 \text{ in } 384,$$

and the estimated probability of high-level professional soccer is

$$\hat{\mathbb{P}}(Y^{F,elite} = 1) = 0.088\% \approx 1 \text{ in } 1,136.$$

Therefore,

$$\frac{\hat{\mathbb{P}}(Y^A = 1)}{\hat{\mathbb{P}}(Y^{F,elite} = 1)} \approx 123.2.$$

Outcome	Estimate	95% CI low	95% CI high
Elite study success	10.839%	10.728%	10.951%
Any pro soccer contract	0.260%	0.242%	0.279%
High-level pro soccer	0.088%	0.077%	0.099%

7.2 Stage-gate funnel

Stage	Count	Probability	One-in
Academic: foundation	203,465	67.822%	1 in 1.5
Academic: strong secondary	132,644	44.215%	1 in 2.3
Academic: elite entry	36,800	12.267%	1 in 8.2
Academic: elite study success	32,518	10.839%	1 in 9.2
Soccer: regional/academy selection	7,643	2.548%	1 in 39
Soccer: academy retention	3,185	1.062%	1 in 94
Soccer: any pro contract	781	0.260%	1 in 384
Soccer: high-level pro	264	0.088%	1 in 1,136

Table 2: Exact stage counts and rates from the frozen simulation output.

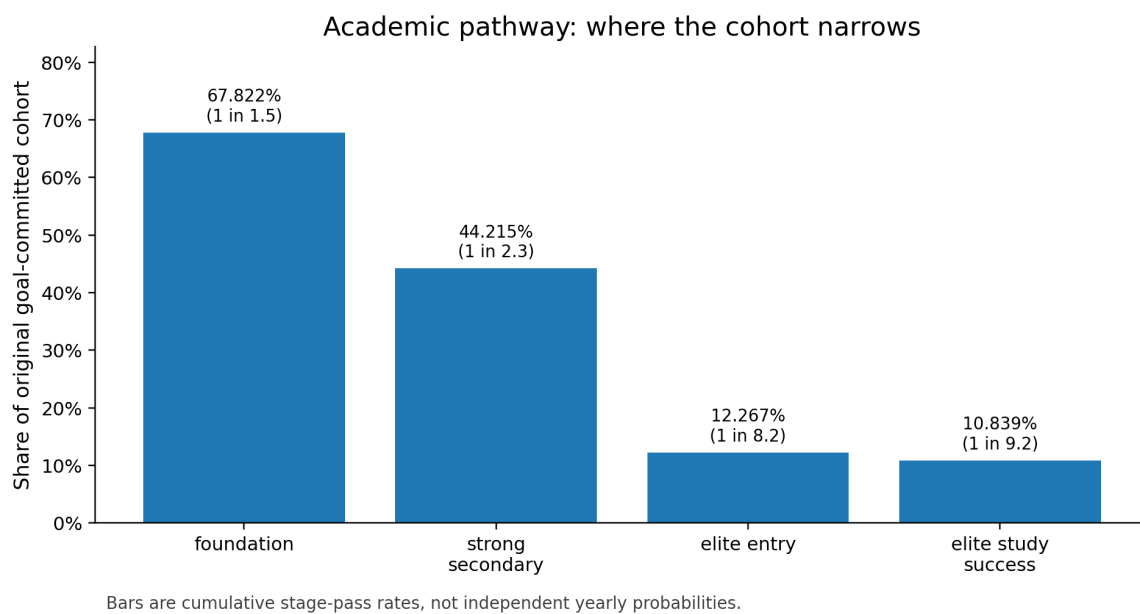


Figure 2: Academic path funnel. The main bottleneck is elite entry, not basic foundation.

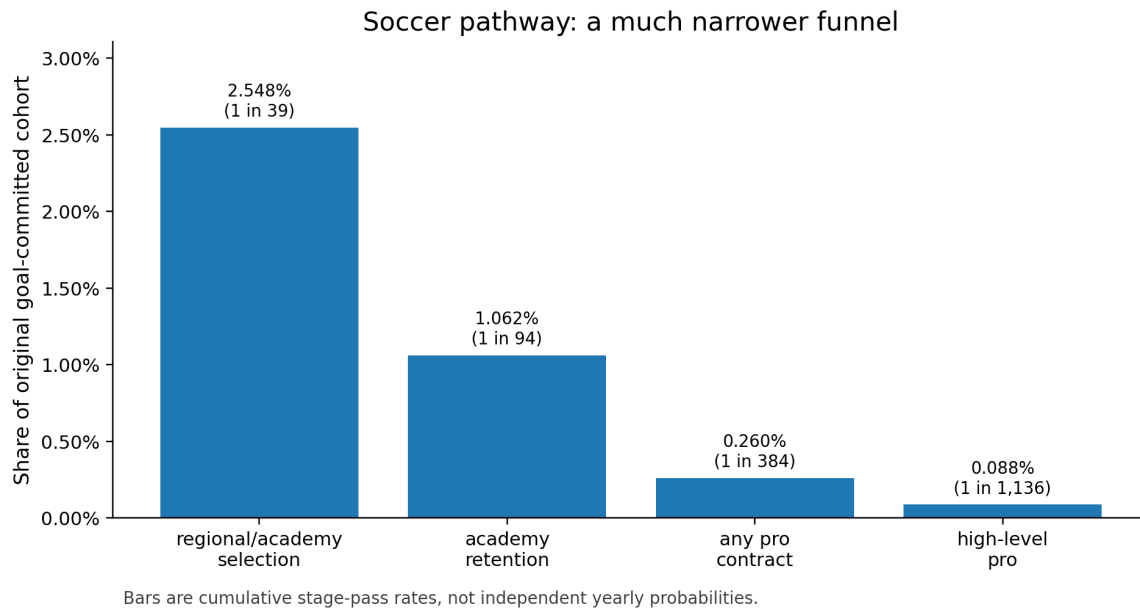


Figure 3: Football path funnel. The path narrows very early and becomes extremely tight at the professional-contract and high-level gates.

7.3 Circumstance sensitivity

The average goal-committed child is not the same as a child already selected into a serious academy, a child in the top academic decile, or a high-athletic child with repeated injuries. Conditioning changes the answer.

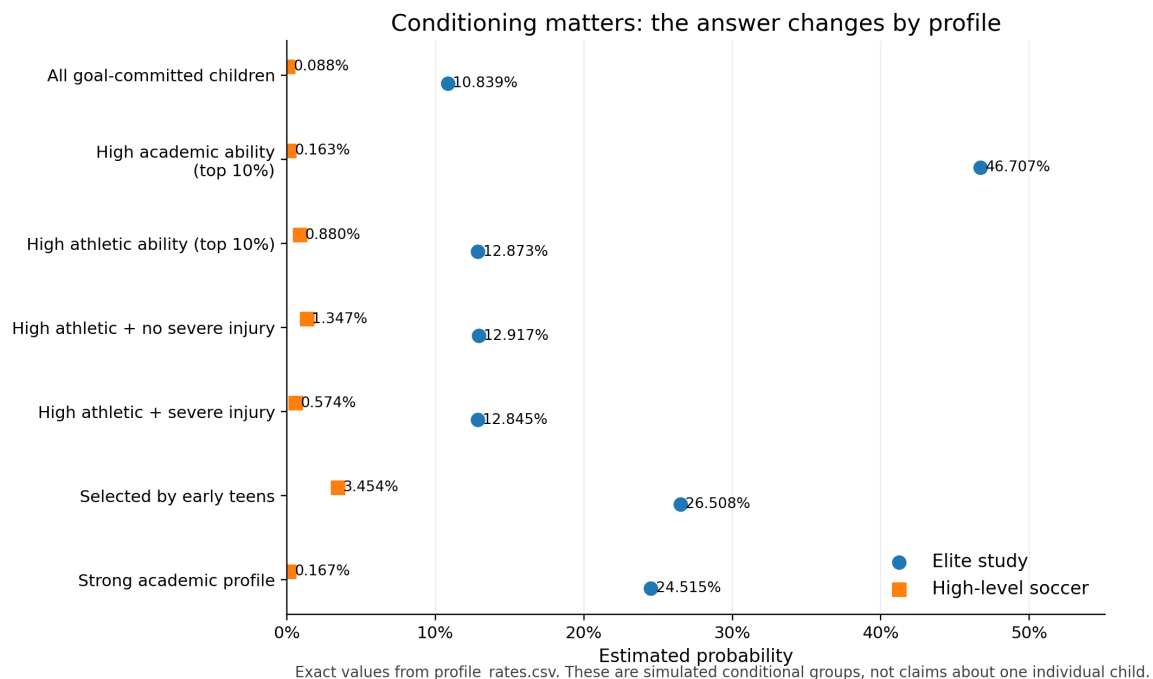


Figure 4: Selected conditional profiles from the exact simulation output. The model supports nuance: football probabilities rise sharply after serious early selection, but the funnel remains narrow.

Profile	N	Elite study	Any pro	High-level
All goal-committed children	300,000	10.839%	0.260%	0.088%

Low current academic ability (bottom 10%)	30,000	0.090%	0.137%	0.037%
High current academic ability (top 10%)	30,000	46.707%	0.433%	0.163%
Low athletic ability (bottom 10%)	30,000	8.687%	0.000%	0.000%
High athletic ability (top 10%)	30,000	12.873%	2.577%	0.880%
Low family resources (bottom 25%)	75,000	1.877%	0.103%	0.035%
High family resources (top 25%)	75,000	24.883%	0.513%	0.173%
High discipline (top 10%)	30,000	35.647%	1.173%	0.477%
Low academic + high athletic	18,781	0.538%	0.751%	0.240%
High academic + low athletic	18,795	27.922%	0.000%	0.000%
High athletic + no severe injury	11,876	12.917%	3.663%	1.347%
High athletic + at least one severe injury	18,124	12.845%	1.865%	0.574%
At least one severe football injury	165,622	10.894%	0.206%	0.063%
No severe football injury	134,378	10.772%	0.327%	0.119%
Soccer-selected by early teens	7,643	26.508%	10.219%	3.454%
Strong academic secondary profile	132,644	24.515%	0.462%	0.167%

Ratio note. The default study/high-level football ratio is 123.2x. For selected groups, the ratio is 14.6x among the top athletic decile, 7.7x among children selected by early teens, and 146.5x among those with a strong academic secondary profile.

Key readings:

- A child in the top 10% of current academic ability reaches elite-study success in 46.707%, while high-level soccer remains 0.163% for that subgroup.
- A child in the top 10% of athletic ability reaches high-level soccer in 0.880%. That is far above the cohort average, but still only about 1 in 114.
- Among children already soccer-selected by early teens, high-level soccer rises to 3.454%, about 1 in 29. This is the clearest evidence that conditioning matters.
- Among children with a strong academic secondary profile, elite-study success is 24.515%, about 1 in 4.1.
- For high-athletic children, severe injury materially changes the football outcome: high-level soccer is 1.347% without severe injury versus 0.574% with at least one severe injury.

7.4 Decile view

The two decile plots below are kept from the simulation output because they show the shape of the risk curve. Academic success increases gradually across academic deciles, then sharply at the top. High-level football is almost invisible until the highest athletic decile because the football gate is much more extreme.

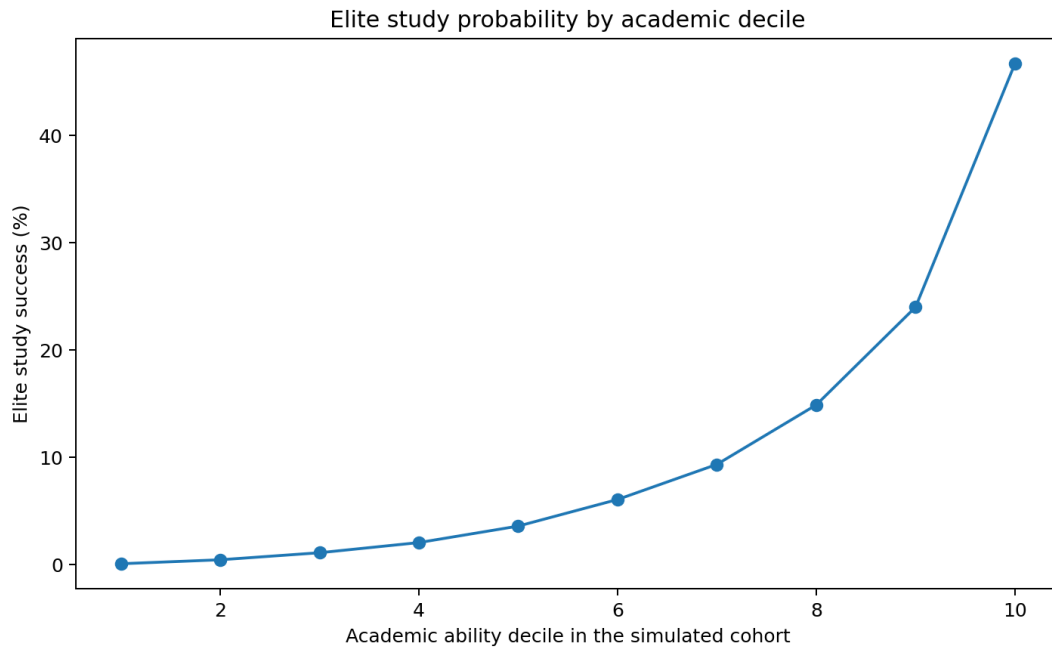


Figure 5: Elite study probability by academic ability decile in the simulated cohort.

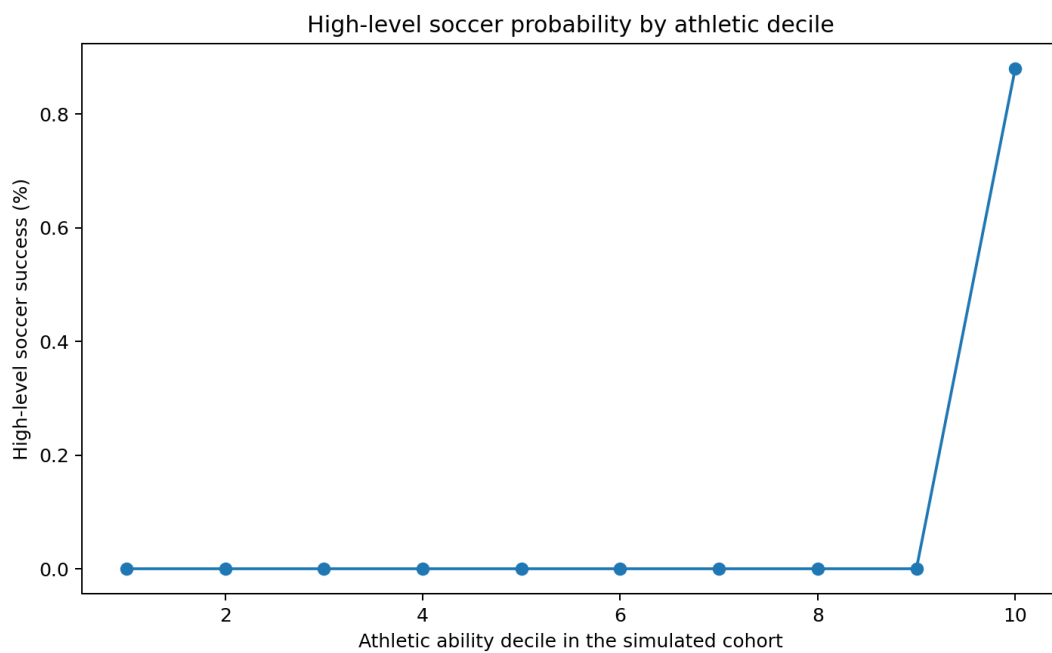


Figure 6: High-level soccer probability by athletic ability decile in the simulated cohort. The visual jump at the last decile reflects the extreme selectivity of the final football gates.

8 Machine-learning interpretability check

The random forest is not the source of the probabilities. The stage-gate model generates the outcomes. The random forest only asks: after the simulation is generated, do the important variables look sensible?

Do not overread the AUC. The high AUC values are not proof that the real world has been solved. They mainly show that the simulated outcomes are predictable from the variables that the model used to generate them. This is an internal coherence check, not external validation.

Audit summary.

- **Elite study outcome:** AUC = 0.936. The strongest audit predictors are tactical learning, academic ability, discipline, family resources, and environment quality.
- **Any professional soccer contract:** AUC = 0.996. The strongest audit predictors are athletic ability, physical profile, health resilience, discipline, and environment quality.
- **High-level professional soccer:** AUC = 0.954. The strongest audit predictors are athletic ability, physical profile, health resilience, tactical learning, and environment quality.

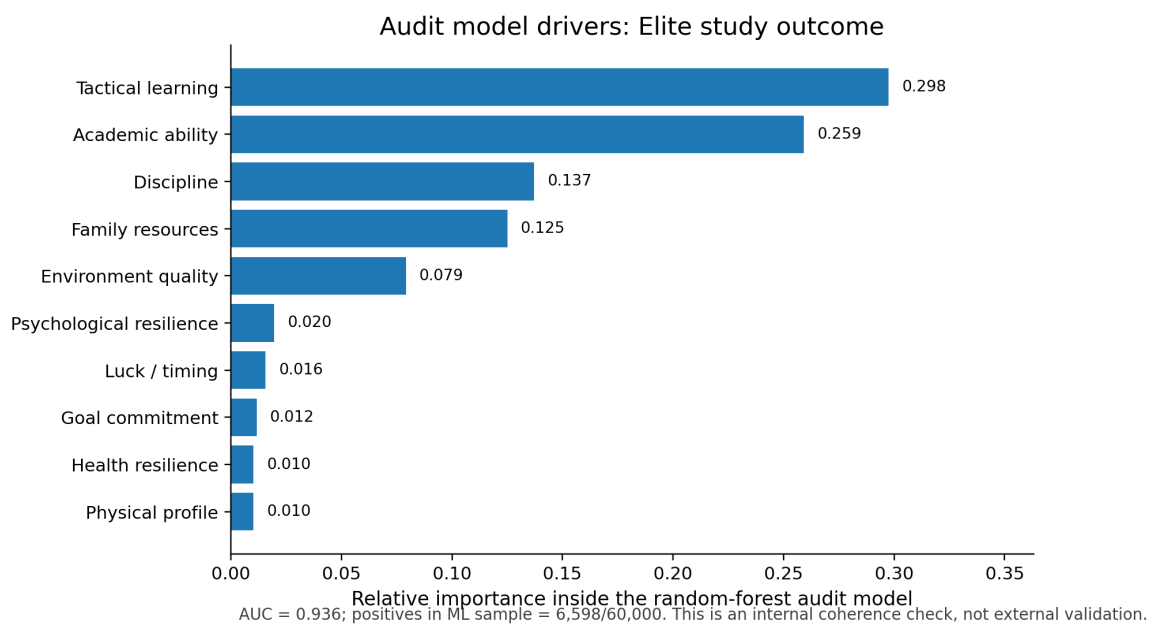


Figure 7: Random-forest audit drivers for the elite study outcome.

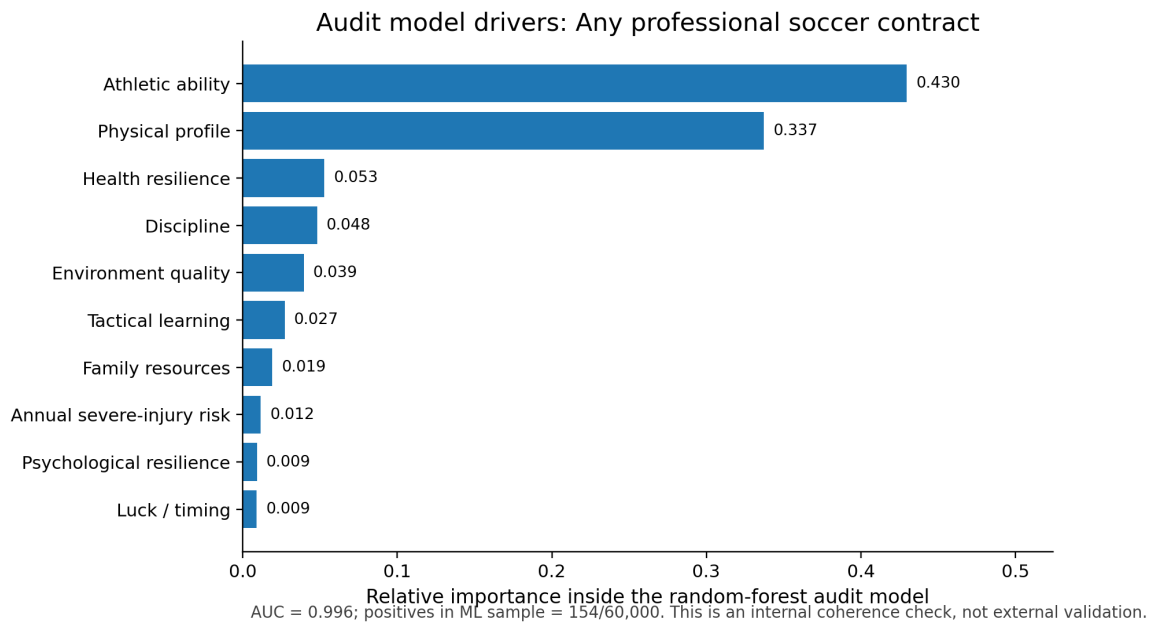


Figure 8: Random-forest audit drivers for any professional soccer contract.

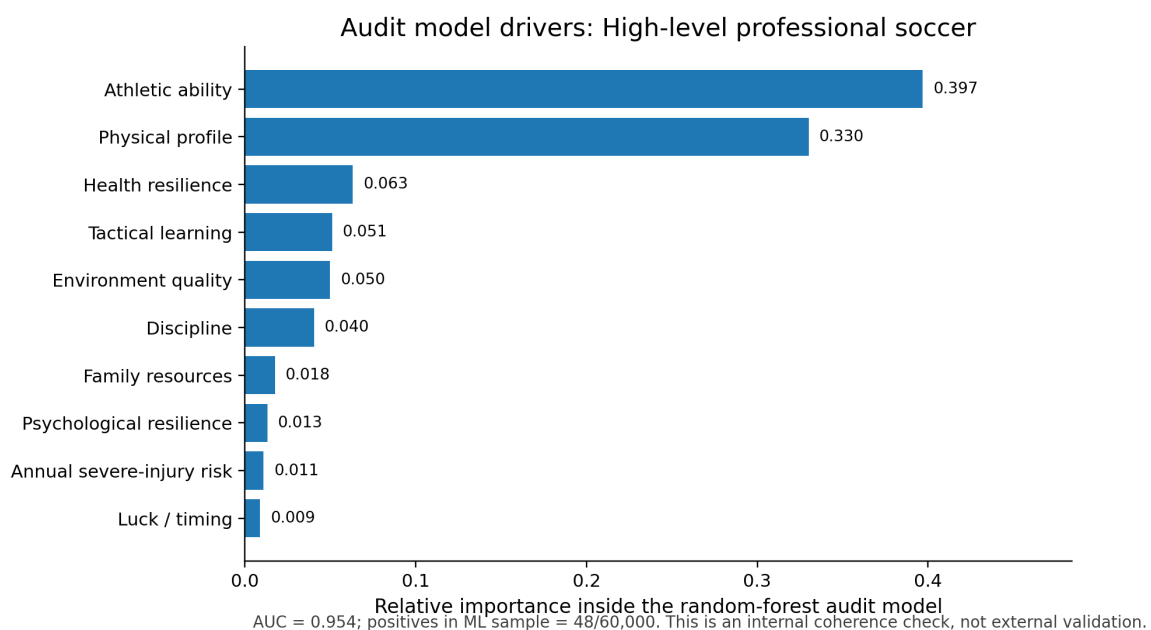


Figure 9: Random-forest audit drivers for high-level professional soccer.

The feature rankings make sense. Elite study is driven by academic ability, discipline, family resources, environment, and the derived tactical-learning variable. Football is driven mainly by athletic ability and physical profile, with health, tactical learning, environment, and discipline behind them.

9 Interpretation rules

9.1 Do not read the result as destiny

A 0.088% probability does not mean a specific child has no chance. It means that, inside a broad goal-committed cohort, high-level professional football is extremely selective. Individual scouting, unusual

talent, exceptional coaching, family sacrifice, and luck can shift a child's conditional probability.

9.2 Conditioning matters

The question “What is the chance from age six?” is very different from “What is the chance after selection into a serious academy at age thirteen?” Mathematically,

$$\mathbb{P}(Y^{F,elite} = 1 \mid G = 1)$$

is much smaller than

$$\mathbb{P}(Y^{F,elite} = 1 \mid G = 1, \text{ academy-selected at 13}).$$

The same logic applies to school. A child who is already a top academic performer at 15 has a different probability from the average goal-committed child.

9.3 Low academic readiness is not low life potential

A child weak at school at age ten is not doomed. But if the target is doctor, elite engineer, or elite technical professional, the model correctly penalizes low current academic readiness unless it improves. In real life, the answer is not shame. The answer is training: sleep, reading, numeracy, tutoring, screen control, stable routines, and better feedback.

على قدر أهل العزم تأتي العزائم
وتأتي على قدر الكرام المكارم

Great ambitions come according to the greatness of determined people; noble deeds come according to the nobility of noble people.

9.4 Injury changes football more than school

A serious knee, ankle, hamstring, or recurrent muscle injury directly attacks the football production function. It removes training time, reduces speed or confidence, and can lower selection probability. Academic routes also suffer from health shocks, but a broken season in football is usually more damaging than a broken semester in school because the football selection window is narrow.

10 What would change the conclusion?

This section is important because a credible model should say what could weaken it.

The conclusion would become less strong if all three of these were true at the same time:

1. the academic path were made much more selective than the current calibration;
2. the football path were conditioned on already-validated elite academy selection, not just early commitment;
3. injury exposure and late-stage football competition were assumed to be much less damaging than the current model assumes.

Even then, the conclusion would not automatically reverse. In the current run, football improves sharply after early selection, but high-level football among soccer-selected children is still 3.454%

while elite study for that subgroup is 26.508%. The model therefore supports a balanced rule: **only overweight football after external football validation; until then, protect school.**

11 Final conclusion

The final calibrated simulation gives a clear decision result. For a young, goal-committed person, the elite-study pathway is statistically far more reliable than the high-level professional football pathway.

$$\hat{\mathbb{P}}(\text{elite study}) = 10.839\% \quad \text{versus} \quad \hat{\mathbb{P}}(\text{high-level football}) = 0.088\%.$$

That is a ratio of about **123.2 to 1** in favor of the elite-study pathway in the default simulation.

وما نيل المطالب بالتمني
ولكن تؤخذ الدنيا غلابا

Great goals are not won by wishing; the world is taken through struggle.

This does not kill ambition. It disciplines ambition. The rational plan is not to forbid football; it is to avoid letting football destroy the academic base before football has produced serious external evidence. Train hard if the talent is real. Compete hard if the opportunity is real. But protect school until the football signal is strong enough to justify the risk.

12 Limitations

1. The model is synthetic. It is calibrated to public anchors but does not follow real children longitudinally.
2. The academic route differs sharply by country. Medicine, engineering, and elite technical tracks do not have identical structures across France, Morocco, Germany, the United States, or other systems.
3. The football route differs by country, academy quality, passport/work-permit constraints, club network, and family mobility.
4. Some variables are hard to measure: discipline, motivation, coach quality, tactical learning, psychological resilience, and luck.
5. The feature-importance charts are internal audit tools, not proof of external predictive validity.
6. The default model is intentionally editable. The strongest use is to change assumptions, rerun, and compare.

A Default parameterization

The default code uses strong negative intercepts for football’s late gates because high-level football is a rare labor-market outcome. It uses less negative intercepts for the academic route because elite study is difficult but has more seats and more parallel branches.

A.1 Correlation assumptions for latent traits

The base latent vector (C, A, D, R, E, H, P, L) is generated from a multivariate normal distribution. Most correlations are zero except the following modelling assumptions:

Pair	Correlation	Reason
$\rho(C, R)$	0.25	Resources often improve academic preparation and stability.
$\rho(C, D)$	0.20	Stronger students often have better study habits, but imperfectly.
$\rho(C, E)$	0.20	Better schools and environments raise measured readiness.
$\rho(R, E)$	0.45	Resources strongly affect school, club, transport, and coaching access.
$\rho(R, D)$	0.15	Stable support can help discipline.
$\rho(A, H)$	0.25	Athletic ability and durability are related but distinct.
$\rho(A, D)$	0.10	Athletic development benefits from discipline.
$\rho(A, E)$	0.10	Good sporting environments reveal and develop ability.
$\rho(D, P)$	0.25	Discipline and psychological resilience often reinforce each other.
$\rho(H, P)$	0.15	Health and resilience interact through recovery and stress.
$\rho(C, L)$	0.05	Luck can slightly affect academic opportunities.
$\rho(A, L)$	0.05	Luck can slightly affect football visibility.

A.2 Derived variables and shocks

$$G_i \sim \text{clip}(\mathcal{N}(1.00, 0.35^2), -0.25, 2.25), \quad (4)$$

$$Q_i = 0.75A_i + 0.25H_i + \epsilon_i^Q, \quad \epsilon_i^Q \sim \mathcal{N}(0, 0.20^2), \quad (5)$$

$$T_i = 0.35C_i + 0.45D_i + 0.20P_i + \epsilon_i^T, \quad \epsilon_i^T \sim \mathcal{N}(0, 0.20^2). \quad (6)$$

The severe-injury probability is

$$p_i^{inj} = \sigma(-2.20 - 0.55H_i + 0.25A_i + 0.15G_i + \epsilon_i^{inj}), \quad \epsilon_i^{inj} \sim \mathcal{N}(0, 0.25^2),$$

with

$$I_i \sim \text{Poisson}(7p_i^{inj}).$$

The career-limiting injury probability is

$$p_i^{end} = \text{clip}(0.015 + 0.045I_i + 0.030 \max(-H_i, 0), 0, 0.60), \quad K_i \sim \text{Bernoulli}(p_i^{end}).$$

The academic disruption probability is

$$p_i^{disrupt} = \sigma(-2.50 - 0.35R_i - 0.30P_i + \epsilon_i^{disrupt}), \quad \epsilon_i^{disrupt} \sim \mathcal{N}(0, 0.25^2),$$

with

$$M_i \sim \text{Bernoulli}(p_i^{disrupt}).$$

A.3 Academic coefficients

$$z_{foundation,i}^A = 0.80 + 1.00C_i + 0.60D_i + 0.60R_i + 0.45E_i + 0.40G_i + 0.20H_i + 0.15L_i - 0.60M_i, \quad (7)$$

$$z_{strong,i}^A = -0.10 + 1.35C_i + 0.90D_i + 0.55R_i + 0.45E_i + 0.35G_i + 0.15H_i + 0.20L_i - 0.70M_i, \quad (8)$$

$$z_{entry,i}^A = -3.50 + 1.50C_i + 1.10D_i + 0.65R_i + 0.55E_i + 0.30G_i + 0.20L_i - 0.60M_i, \quad (9)$$

$$z_{completion,i}^A = -0.20 + 0.85C_i + 1.15D_i + 0.45R_i + 0.35E_i + 0.25G_i + 0.35H_i + 0.20P_i + 0.25L_i - 0.80M_i. \quad (10)$$

A.4 Football coefficients

$$z_{selection,i}^F = -7.50 + 1.90A_i + 0.80Q_i + 0.65D_i + 0.45R_i + 0.65E_i + 0.55G_i + 0.30H_i + 0.20RA_i + 0.25L_i - 0.30K_i, \quad (11)$$

$$z_{retention,i}^F = -7.50 + 2.10A_i + 0.95Q_i + 0.90D_i + 0.35R_i + 0.75E_i + 0.45G_i + 0.35H_i + 0.25RA_i + 0.20L_i - 0.35I_i - 2.00K_i, \quad (12)$$

$$z_{contract,i}^F = -10.50 + 2.30A_i + 1.10Q_i + 1.00D_i + 0.25R_i + 0.70E_i + 0.35G_i + 0.30T_i + 0.25L_i - 0.65I_i - 2.00K_i, \quad (13)$$

$$z_{high,i}^F = -12.00 + 2.40A_i + 1.20Q_i + 1.10D_i + 0.20R_i + 0.70E_i + 0.30G_i + 0.35T_i + 0.30L_i - 0.85I_i - 2.50K_i. \quad (14)$$

Here RA_i is relative-age advantage. It affects early selection and retention but not adult professional success.

B Mathematical summary

$$X_i \sim \mathcal{N}(0, \Sigma), \quad (15)$$

$$p_{d,t,i} = \sigma(\alpha_{d,t} + \beta_{d,t}^\top X_i - \Delta_{d,t,i}), \quad (16)$$

$$B_{d,t,i} \sim \text{Bernoulli}(p_{d,t,i}), \quad (17)$$

$$Y_{d,i} = \prod_t B_{d,t,i}, \quad (18)$$

$$\hat{\mathbb{P}}(Y_d = 1) = \frac{1}{N} \sum_i Y_{d,i}, \quad (19)$$

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}, \quad (20)$$

$$CI_{95\%} = \hat{p} \pm 1.96SE. \quad (21)$$

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